Thurrje - Sinjalet Diskrete

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

"$n \to k$"
"$-k$"
"$n-k$"

$$y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k)$$

Formula

I $$\sum_{k=0}^{\infty} \lambda^k = \frac{1}{1-\lambda}$$

Sh. $$\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{1-\frac{1}{2}} = 2$$

II $$\sum_{k=0}^{N} \lambda^k = \frac{1-\lambda^{N+1}}{1-\lambda}$$

Sh. $$\sum_{k=0}^{7} 2^k = \frac{1-2^{7+1}}{1-2} = \frac{1-2^8}{1-2}$$

III $$\sum_{k=0}^{N} \lambda^k = \sum_{k=0}^{N} 1 = N+1$$

Sh. $$\sum_{k=0}^{6} 1 = 6+1 = 7$$

2.28

Janë dhënë hyrja $x(n)$ dhe përgjigjja impulsive $h(n)$ si në vijim:

$$x(n) = u(n)$$

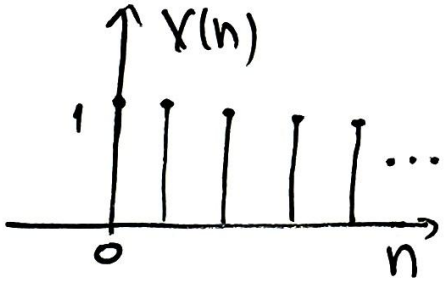
$$h(n) = \alpha^n u(n) \qquad 0<\alpha<1 \quad \text{p.sh. } \alpha = \frac{1}{2}$$

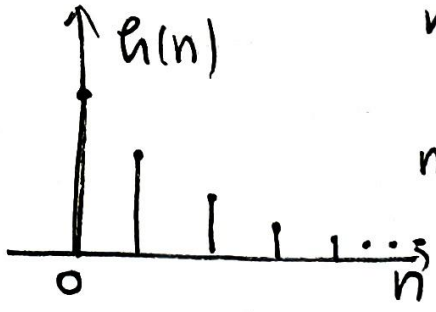
$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

a) $y(n) = ?$ $\quad y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k)$

b) $y(n) = ?$ $\quad y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k)\,x(n-k)$

Zgjidhje



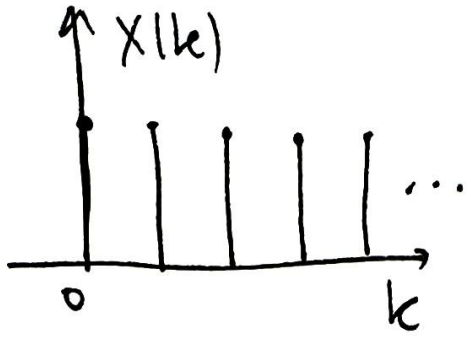


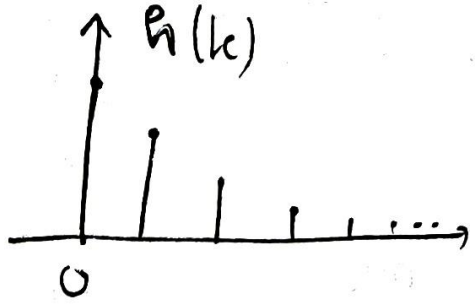
$n=0,\ \left(\frac{1}{2}\right)^0 = 1$

$n=1;\ \left(\frac{1}{2}\right)^1 = 0.5$

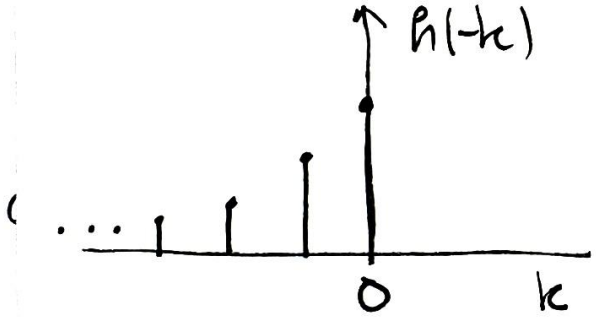
$n=2;\ \left(\frac{1}{2}\right)^2 = 0.25$

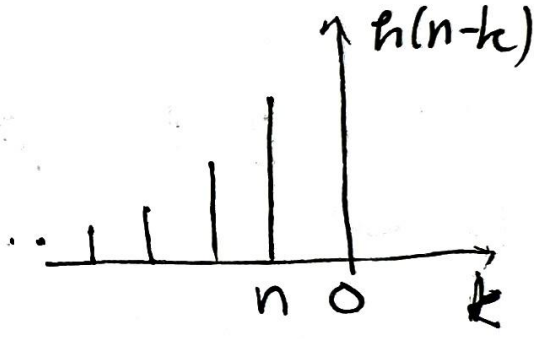
1° Kalimi në "k"



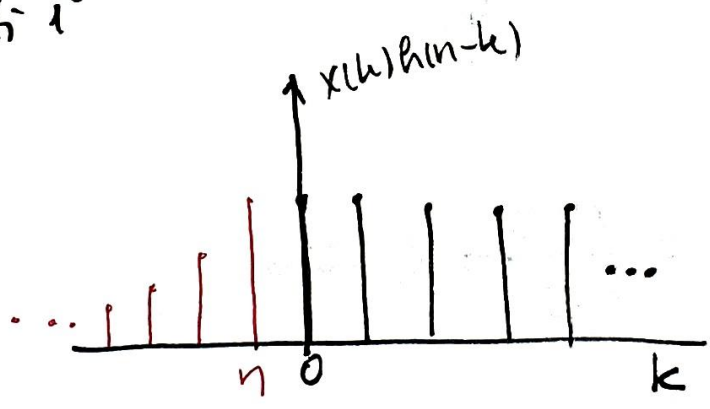


: a) Pasqyrojmë dhe zhvendosim h(k)





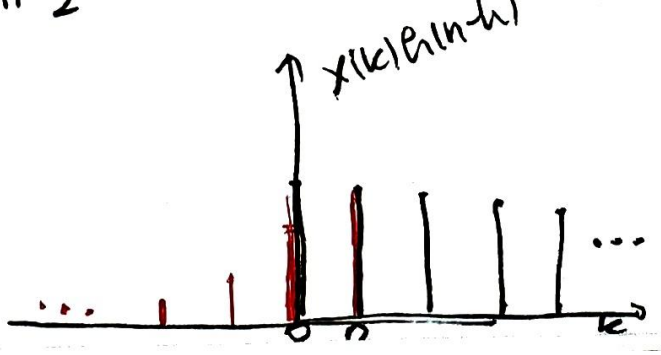
① Rasti 1°



1° $n<0$

$y(n)=0$

Rasti 2°



2° $n>0$

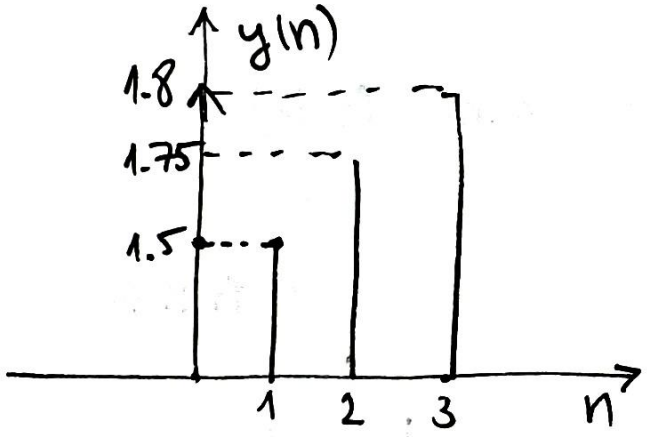
$$y(n)=\sum_{k=0}^{n} 1\cdot\left(\frac{1}{2}\right)^{n-k}$$

$n-k=m$

$$y(n)=\sum_{m=0}^{n}\left(\frac{1}{2}\right)^{m}=\frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}}$$

Detyra nuk ka nqaste fiera!

$y(n)$



$n>0 \Rightarrow (1, \infty)$

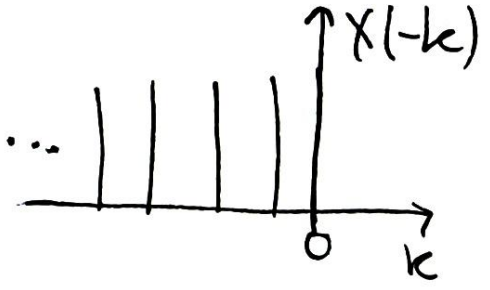
$n=1$, $y(n)=\frac{1-\left(\frac{1}{2}\right)^{1+1}}{1-\frac{1}{2}}=\frac{1-\frac{1}{4}}{1-\frac{1}{2}}=1.5$

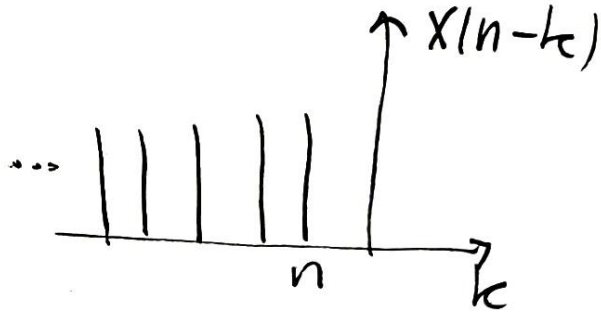
$n=2$ $y(n)=\frac{1-\frac{1}{8}}{1-\frac{1}{2}}=\frac{0,87}{0,5}=1,756$

$n=3$ $y(n)=\frac{1-\frac{1}{16}}{1-\frac{1}{2}}=1,8$

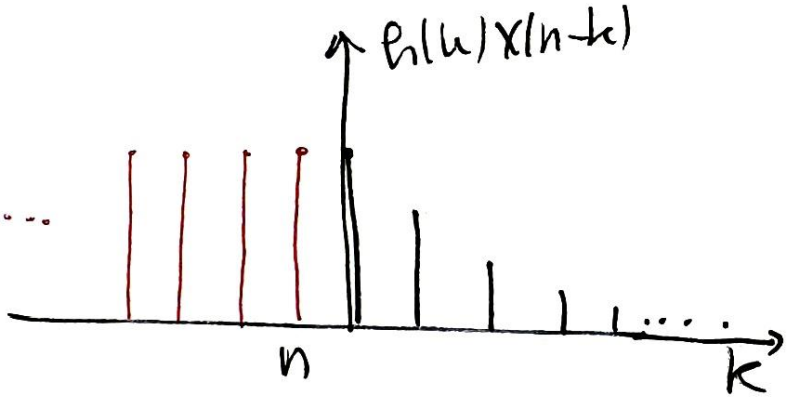
b)

$y(n) = h(n) * x(n)$





Rasti 1°



1° $n < 0$

$y(n) = 0$

Rasti 2°



2° $n > 0$

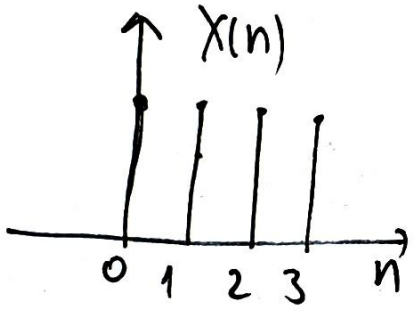
$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\,x(n-k) = \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k \cdot 1 = \frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}}$$

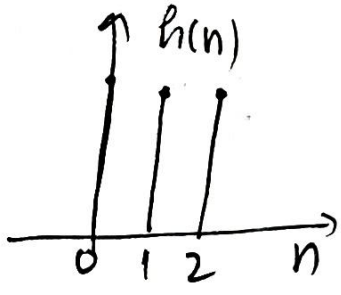
Rezultati i njejtë me rastin a)

2.30 Gjeni dalgen $y(n)$ nëse janë dhënë $x(n)$ dhe $h(n)$

a) M. analitike

b) M. grafikone





Zgjidhje

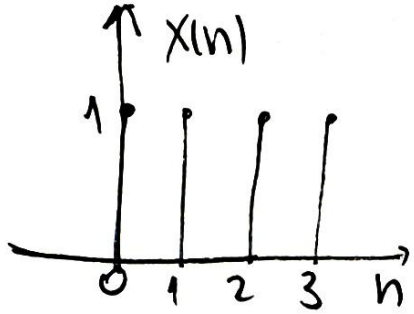
a)

$$y(n) = x(n) * h(n)$$

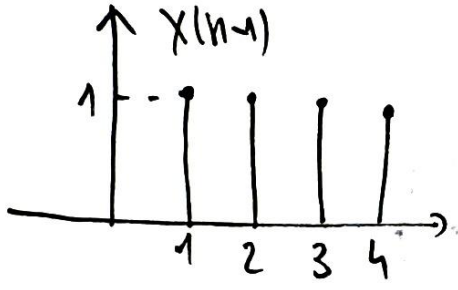
$$h(n) = \delta(n) + \delta(n-1) + \delta(n-2)$$

$$y(n) = x(n) * \{\delta(n) + \delta(n-1) + \delta(n+2)\}$$

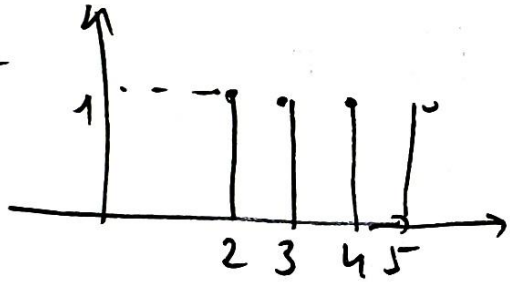
$$= x(n) + x(n-1) + x(n-2).$$



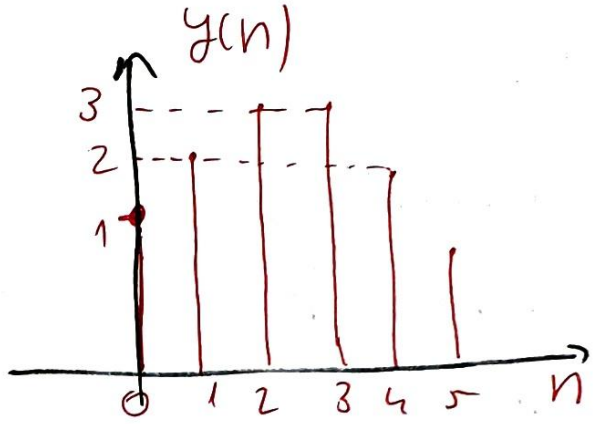
+



+



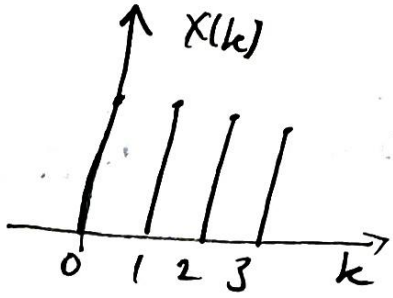
$\Rightarrow$



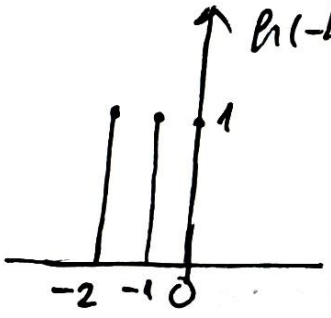
$$y(n) = \delta(n) + 2\delta(n-1) + 3\delta(n-2)$$
$$+ 3\delta(n-3) + 2\delta(n-4) + \delta(n-5)$$

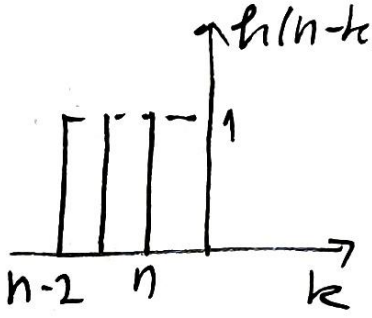
b) M. Grafikone



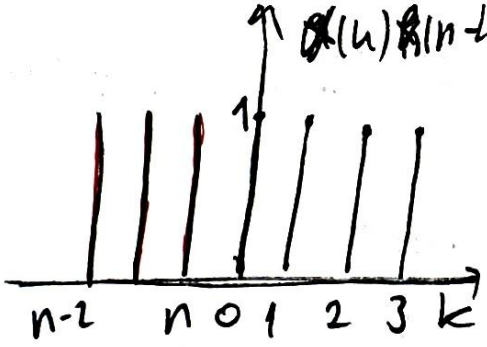


7





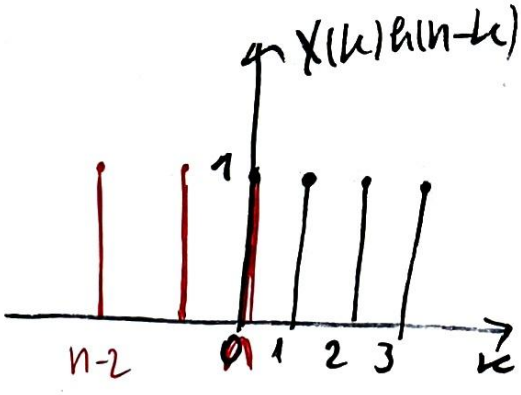
①



① $n < 0$

$y(n) = 0$

②

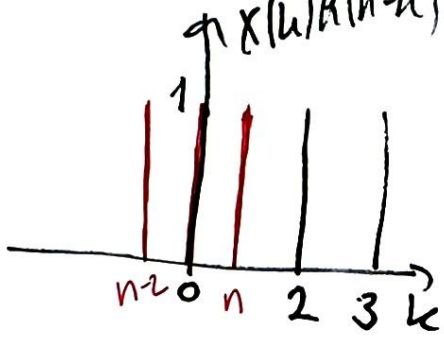


② $n = 0$

$$y(n) = \sum_{k=0}^{0} x(k)\,h(n-k)$$

$$y(n) = x(0)\,h(0-0) = 1 \cdot 1 = 1$$

3.



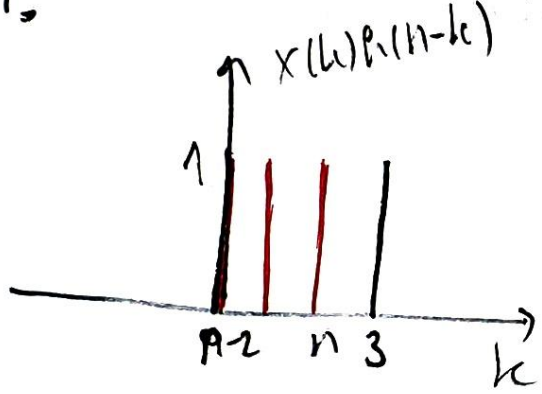
3. $n=1$

$$y(n)=\sum_{k=0}^{1} x(k)h(n-k)$$

$$y(n)= x(0)x(1-0)+x(1)h(1-1)$$

$$= 1\cdot 1 + 1\cdot 1 = 2$$

4.



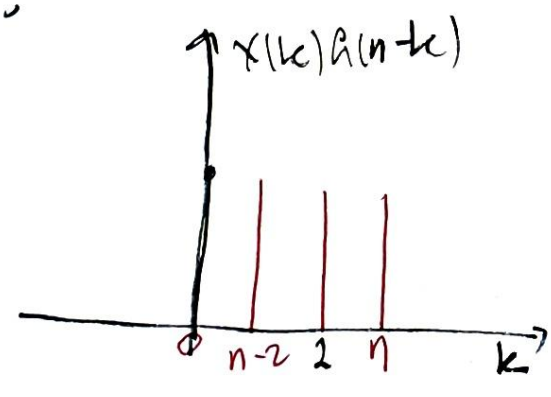
4. $n=2$

$$y(n)=\sum_{k=0}^{2} x(k)h(n-k)$$

$$y(n)= x(0)h(2-0)+x(1)h(2-1)$$

$$+x(2)h(2-2)= 1+1+1=3$$

5.



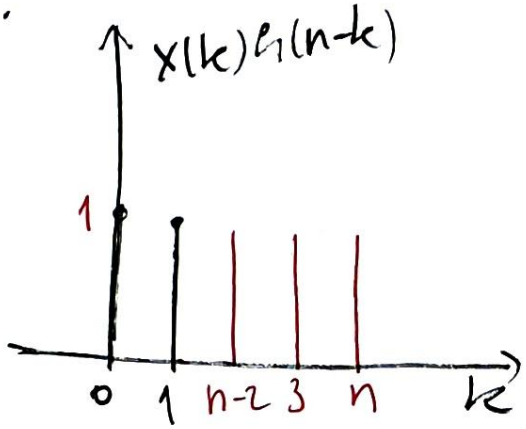
5. $n=3$

$$y(n)=\sum_{k=1}^{3} x(k)h(n-k)=$$

$$x(1)h(3-1)+x(2)h(3-2)$$

$$+x(3)h(3-3)=1+1+1=3$$

6.



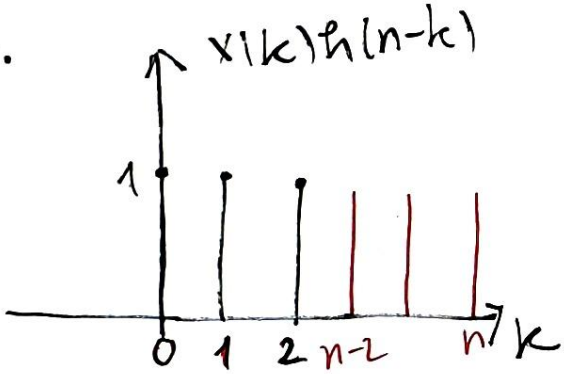
6. $n=4$

$$y(n)=\sum_{k=2}^{3} x(k)h(n-k)$$

$$= x(2)h(4-2)+x(3)h(4-3)$$

$$= 1+1=2$$

7.

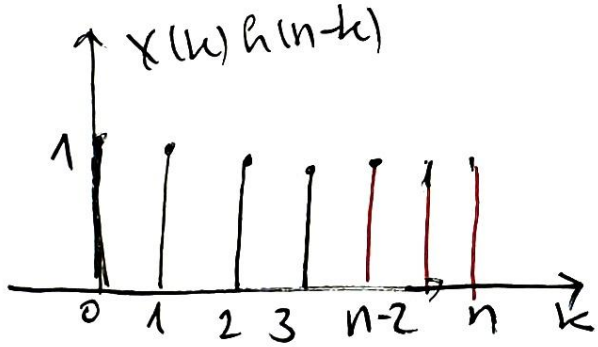


7. $n=5$

$$y(n)=\sum_{k=3}^{3} x(k)h(n-k)$$

$$= x(3)h(5-3) = 1\cdot 1 = 1$$

8.



8. $n>5$

$y(n)=0$

Figura per $y(n)$

